Assignment 3

Name – Kale Sushant Adhirath

RollNo. – 96705

Assign 3 : Linux Commands		
Please find the commands for the following. (Find appropriate command line options for the commands and write commands solutions into your notebook)		
No.	**Commands**	**To Do**
1.	ls	Display hidden files
2.	rename	Rename all *.txt files to *.htm
3.	ln -s	Create symbolic link (shortcut) for a file
4.	ln	Create hard link for a file
5.	stat	Display time stamp of the file
6.	touch	Change the time stamp of the file
7.	touch	Create empty file
8.	tail	Print all lines in a file from fifth line onwards
9.	init 0, poweroff	Shutdown (Only for super-user)
10.	init 6, reboot	Reboot (Only for super-user)
12.	whoami	Find current user
13.	tty	Find current terminal
14.	who am i	Find current user and terminal
15.	users	Find all users logged into the system
16.	who	Find all users logged in and terminals in use
17.	grep, egrep, fgrep	Find a given word/pattern from the file
18.	chmod	Make a file as read-only
19.	chmod	Make a file as write, read and execute
20.	wc (using redirection)	Take input from the file, count words and store output to other file
21.	who | wc (using pipe)	Count words in output of "who" command
22.	head, tail (using pipe)	Print lines 5 to 10 from the given file
23.	sort , uniq (using pipe)	Print the uniq values from the unsorted file
24.	who, grep (using pipe)	Display all users not using tty2, tty3, tty4
25.	ls	Display all files in ascending order of their file size
26.	ls, grep (using pipe)	Count number of files and directories from current directory
27.	ls, grep (using pipe)	Display only hidden files in a directory
28.	tr	Convert file contents into uppercase , also learn "-s" option of this command
29.	cut	Split the file contents using delimiter and print required fields

1.ls

```
vagrant@ubuntu-jammy:~/sunbeam$ ls -a
.  ..  18oct  3.txt  33.txt  data.txt  file.txt  grade.txt  one
```

2.rename

```
vagrant@ubuntu-jammy:~/sunbeam$ ls -a
.  ..  18oct  3.txt  33.txt  data.txt  file.txt  grade.txt  one
vagrant@ubuntu-jammy:~/sunbeam$ mv file.txt doc.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  doc.txt  grade.txt  one
vagrant@ubuntu-jammy:~/sunbeam$
```

3.ln -s

```
vagrant@ubuntu-jammy:~/sunbeam$ ln -s target linkname
```

4.ln

```
vagrant@ubuntu-jammy:~/sunbeam$ ln file1 file2
```

5.stat

```
vagrant@ubuntu-jammy:~/sunbeam$ stat doc.txt
  File: doc.txt
  Size: 2850            Blocks: 8          IO Block: 4096   regular file
Device: 801h/2049d      Inode: 256259      Links: 1
Access: (0664/-rw-rw-r--)  Uid: ( 1000/ vagrant)   Gid: ( 1000/ vagrant)
Access: 2026-03-24 16:03:55.219378296 +0000
Modify: 2026-03-24 16:03:51.879260870 +0000
Change: 2026-03-24 16:18:54.791817384 +0000
 Birth: 2026-03-24 16:03:51.879260870 +0000
```

6.touch

```
vagrant@ubuntu-jammy:~/sunbeam$ touch file.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  file.txt  grade.txt  one
```

8.tail

```
vagrant@ubuntu-jammy:~/sunbeam$ touch file.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls
18oct  3.txt  33.txt  data.txt  file.txt  grade.txt  one
```

9.int 0,power off

```
vagrant@ubuntu-jammy:~/sunbeam$ init 0
poweroff
shutdown -h now
```

10.init 6,reoboot

```
vagrant@ubuntu-jammy:~/sunbeam$ init 6
reboot
shutdown -r now
```

12.whoami

```
vagrant@ubuntu-jammy:~/sunbeam$ whoami
vagrant
```

13.tty

```
vagrant@ubuntu-jammy:~/sunbeam$ tty
/dev/pts/0
```

14.who am i

```
vagrant@ubuntu-jammy:~/sunbeam$ who am i
vagrant  pts/0        2026-03-24 13:17 (10.0.2.2)
```

15.users

```
vagrant@ubuntu-jammy:~/sunbeam$ users
vagrant
```

16.who

```
vagrant@ubuntu-jammy:~/sunbeam$ who
vagrant  pts/0        2026-03-24 13:17 (10.0.2.2)
```

17.grep,egrep,fgrep

```
vagrant@ubuntu-jammy:~/sunbeam$ grep "pattern" file.txt
egrep "pattern" file.txt
fgrep "pattern" file.txt
```

18.chmod - Make a file as read-only

```
vagrant@ubuntu-jammy:~/sunbeam$ chmod 444 doc.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls -l
total 28
drwxrwxr-x 5 vagrant vagrant 4096 Mar 24 12:57 18oct
-rw-rw-r-- 1 vagrant vagrant   21 Mar 24 15:53 3.txt
-rw-rw-r-- 1 vagrant vagrant    7 Mar 24 15:49 33.txt
-rw-rw-r-- 1 vagrant vagrant  145 Mar 24 08:51 data.txt
-r--r--r-- 1 vagrant vagrant 2850 Mar 24 16:03 doc.txt
-rw-rw-r-- 1 vagrant vagrant  153 Mar 24 11:55 grade.txt
lrwxrwxrwx 1 vagrant vagrant    6 Mar 24 17:07 linkname -> target
drwxrwxr-x 3 vagrant vagrant 4096 Mar 24 15:20 one
```

19.chmod - Make a file as write, read and execute

```
vagrant@ubuntu-jammy:~/sunbeam$ chmod 777 doc.txt
vagrant@ubuntu-jammy:~/sunbeam$ ls -l
total 28
drwxrwxr-x 5 vagrant vagrant 4096 Mar 24 12:57 18oct
-rw-rw-r-- 1 vagrant vagrant   21 Mar 24 15:53 3.txt
-rw-rw-r-- 1 vagrant vagrant    7 Mar 24 15:49 33.txt
-rw-rw-r-- 1 vagrant vagrant  145 Mar 24 08:51 data.txt
-rwxrwxrwx 1 vagrant vagrant 2850 Mar 24 16:03 doc.txt
-rw-rw-r-- 1 vagrant vagrant  153 Mar 24 11:55 grade.txt
lrwxrwxrwx 1 vagrant vagrant    6 Mar 24 17:07 linkname -> target
drwxrwxr-x 3 vagrant vagrant 4096 Mar 24 15:20 one
```

20. wc (using redirection) Take input from the file, count words and store output to other file

```
vagrant@ubuntu-jammy:~/sunbeam$ wc -w < doc.txt > output.txt
```

21. who | wc (using pipe) Count words in output of “who” command

```
vagrant@ubuntu-jammy:~/sunbeam$ who | wc -w
5
```

22. head, tail (using pipe) Print lines 5 to 10 from the given file for doc.txt

```
vagrant@ubuntu-jammy:~/sunbeam$ head -n 10 doc.txt | tail -n 6
bc < input.txt 1> output.txt 2> error.txt
cat output.txt
cat error.txt
sort numbers.txt
sort numbers.txt > sorted.txt
cat sorted.txt
```

23. sort , uniq (using pipe) Print the uniq values from the unsorted file

```
vagrant@ubuntu-jammy:~/sunbeam$ sort doc.txt | uniq
Linux is a free, open-source, Unix-like operating system kernel first releas
ed by Linus Torvalds in 1991, forming the basis for diverse operating system
s known as distributions (distros). It is renowned for its security, stabili
ty, and versatility, powering the vast majority of web servers, supercompute
rs, and cloud infrastructure, alongside Android devices and embedded systems
.
```

24. who, grep (using pipe) - Display all users not using tty2, tty3, tty4

```
vagrant@ubuntu-jammy:~/sunbeam$ who | grep -v -E "tty2|tty3|tty4"
vagrant  pts/0        2026-03-24 13:17 (10.0.2.2)
```

25. ls -Display all files in ascending order of their file size

```
vagrant@ubuntu-jammy:~/sunbeam$ ls -lS -r
total 32
-rw-rw-r-- 1 vagrant vagrant    4 Mar 24 17:20 output.txt
lrwxrwxrwx 1 vagrant vagrant    6 Mar 24 17:07 linkname -> target
-rw-rw-r-- 1 vagrant vagrant    7 Mar 24 15:49 33.txt
-rw-rw-r-- 1 vagrant vagrant   21 Mar 24 15:53 3.txt
-rw-rw-r-- 1 vagrant vagrant  145 Mar 24 08:51 data.txt
-rw-rw-r-- 1 vagrant vagrant  153 Mar 24 11:55 grade.txt
-rw-rw-r-- 1 vagrant vagrant  382 Mar 24 17:25 doc.txt
drwxrwxr-x 3 vagrant vagrant 4096 Mar 24 15:20 one
drwxrwxr-x 5 vagrant vagrant 4096 Mar 24 12:57 18oct
```

26.ls, grep (using pipe) Count number of files and directories from current directory

```
vagrant@ubuntu-jammy:~/sunbeam$ ls -l | grep -v "^total" | wc -l
9
```

27. ls, grep (using pipe) Display only hidden files in a directory

```
vagrant@ubuntu-jammy:~/sunbeam$ ls -a | grep "^\."
.
..
```

28. tr Convert file contents into uppercase , also learn "-s" option of this command

```
vagrant@ubuntu-jammy:~/sunbeam$ tr -s ' ' < doc.txt
Linux is a free, open-source, Unix-like operating system kernel first releas
ed by Linus Torvalds in 1991, forming the basis for diverse operating system
s known as distributions (distros). It is renowned for its security, stabili
ty, and versatility, powering the vast majority of web servers, supercompute
rs, and cloud infrastructure, alongside Android devices and embedded systems
.
```

29. cut Split the file contents using delimiter and print required fields

```
vagrant@ubuntu-jammy:~/sunbeam$ cut -d ":" -f1 doc.txt
Linux is a free, open-source, Unix-like operating system kernel first releas
ed by Linus Torvalds in 1991, forming the basis for diverse operating system
s known as distributions (distros). It is renowned for its security, stabili
ty, and versatility, powering the vast majority of web servers, supercompute
rs, and cloud infrastructure, alongside Android devices and embedded systems
.
```